



Thursday October 1 2026

Location: Curie B

BrainWorks 2026 offers a forum for advancing pathology-aware, generalizable computational neuroimaging and computational neuropathology methods. Presentations will address pediatric anomaly detection, brain tumor segmentation, automated grading of white matter hyperintensities, and new methods for improving the sensitivity of neuroimaging analyses in the presence of diffuse pathology. Generalization is explored through foundation models and self-supervised learning. Contributions span structural and functional neuroimaging as well as light-sheet microscopy, reflecting the cross-modality, cross-pathology focus of BrainWorks.

For more information, please visit: <https://brainworksmiccai.github.io/>

Schedule:

- **08:00 - 08:05 Opening remarks**
- **08:05 - 08:15: Keynote 1:** Bringing AI into neurotrauma (Matthew Pease, Indiana University School of Medicine, USA)
- **08:15 – 09:51 Oral Presentations (10'+2'):**
 - 8:15-8:27 RBT-Faz: Register-Bottleneck 3D Swin-UNETR with Clinical Text Alignment for Ordinal Fazekas Grading from FLAIR MRI
Farzan Niknejad Mazandarani, Kai La Jennifer Tsang, Nicholas Simard, Mumu Aktar, and April Khademi*
 - 8:27-8:39 Hierarchical Differential MedViT-3D: Specialized Hybrid Transformer for Multiclass Diagnosis of Neurodegenerative Diseases
Éloi Navet, Rémi Giraud, Boris Mansencal, and Pierrick Coupé*
 - 8:39-8:51 Unsupervised Anomaly Detection in Pediatric Brains: Method Comparison Across Multiple Clinical Conditions
Francesca Maccarone, Simone Melzi, Cosmin Ionut Bercea, Filippo Arrigoni, Julia Anne Schnabel, Lina Felsner, and Denis Peruzzo*
 - 8:51-9:03 Scaling Native 3D Self-Supervised DINOv2 for Brain MRI Foundation Models
*Shubham Innani, Carla Pitarch-Abaigar, and Spyridon Bakas**
 - 9:03-9:15 Self-supervised Learning for 3D Light-Sheet Microscopy Image Segmentation based on DINO-v2 and nnU-Net
*Siddhesh Thakur, Sai Prajwal Reddy, Sylwia Malec, Chetan Poudel, Eleftherios Garyfallidis, and Spyridon Bakas**
 - 9:15-9:27 Pre-Operative to Intra-Operative: A Brain Tumor Segmentation Generalization Study
*Danish Ali, Amirali Azimi, Colin Patrick Galvin, Tina Kapur, Alexandra Golby, Ajmal Mian, Naveed Akhtar, and Ghulam Mubashar Hassan**

- 9:27-9:39 CASA-Net: Confidence-Aware Spatial Attention Network for Efficient Detection-Guided Brain Tumor Segmentation
Somayeh Davar and Thomas Fevens*
- 9:39-9:51 Group ICA 2.0: Closing the Gap Between Subjects and Group Latent Decomposition
*Oktay Agcaoglu**

**Presenting author*

- **09:51 - 10:00: Session Conclusion**
- **10:00 - 10:30: Poster Session (during MICCAI COFFEE BREAK SLOT)**
- **10:30 - 11:25: Keynote 2:** Worldwide Collaboration, AI, & Data Scaling Laws in the Quest to Map Human Brain Diseases (Paul Thompson)

BrainWorks Posters:

1. *From Edge to Gradient: Quantifying the Infiltrative Margin in Paediatric Diffuse Glioma*
Jesús Ortiz Perona, Jorge Reig Calatayud, Luis Álvarez Mira, and José Francisco Cánovas Cerezo
2. *BrainMRI-CLIP: Enhancing Brain MRI Representation Learning via Joint Vision-Language-Diagnosis Alignment*
Qiankun Zheng, Xintong Wu, Yaping Wu, Yuxiao Liu, Zhaoyu Qiu, Yichu He, Feng Shi, Meiyun Wang, and Dinggang Shen
3. *Direct Mask-Perturbation Auditing for Geometry-Derived Brain Lesion Biomarkers*
Prateek Mittal, Prabal Pratap Singh, and Joohi Chauhan
4. *Reconstruct and Distill: A Hybrid Masked-Autoencoder and Self-Distillation Foundation Model for Clinical Brain MRI*
Nguyen Quoc Hung, Minh Nguyen Dang, Le Nhu Quynh, and Linh Nguyen
5. *UPStain: Unsupervised Population-Based Morphology-Aware Stain Normalization*
Sanyukta Adap, Bhakti Baheti, and Spyridon Bakas
6. *DeepNucleiNet: Encoding Spatial Nuclear Patterns for Data-Efficient Necrosis Detection in Computational Neuropathology*
Dasari Naga Raju, T K Srikanth, Shilpa Rao, Ramesh Kestur, and Mahadevan A
7. *Bidirectional Propagation Makes Sparse Re-Prompting Viable in Zero-Shot SAM2 for Whole-Tumor Glioma Segmentation*
Sheikh Ziauddin and Mirza Tahir Ahmed
8. *Reliability-aware Multimodal Fusion for Brain Tumor Segmentation*
Xinyue Wang, Yumou Zhou, and Jinyan Fan